Exercise 1.5

1. **Fill in the blanks:**
   (i) The ones digits in the cube of 73 is_____.
   (ii) The maximum number of digits in the cube of a two digit number is _____.
   (iii) The smallest number to be added to 3333 to make it a perfect cube is _____.
   (iv) The cube root of 540×50 is _____.
   (v) The cube root of 0.000004913 is _____.



2. **Say True or False:**
   (i) The cube of 24 ends with the digit 4.
   (ii) Subtracting 10^3 from 1729 gives 9^3.
   (iii) The cube of 0.0012 is 0.000001728.
   (iv) 79570 is not a perfect cube.
   (v) The cube root of 250047 is 63.
3. Show that 1944 is not a perfect cube.
4. Find the smallest number by which 10985 should be divided so that the quotient is a perfect cube.
5. Find the smallest number by which 200 should be multiplied to make it a perfect cube.
6. Find the cube root of $24 \times 36 \times 80 \times 25$.
7. Find the cube root of 729 and 6859 by prime factorisation.
8. What is the square root of cube root of 46656?
9. If the cube of a squared number is 729, find the square root of that number.
10. Find the two smallest perfect square numbers which when multiplied together gives a perfect cube number.



Activity

Observe that

$2^3 - 1^3 = 1 + 2 \times 1 \times 3$

$3^3 - 2^3 = 1 + 3 \times 2 \times 3$

$4^3 - 3^3 = 1 + 4 \times 3 \times 3$

Find the value of $15^3 - 14^3$ in the above pattern.

Observe that

$1^3 = 1 = 1$

$2^3 = 8 = 3 + 5$

$3^3 = 27 = 7 + 9 + 11$

Continue this pattern to find the value of 7^3 as the sum of consecutive odd numbers.

1.9 Exponents and Powers

We know how to express some numbers as squares and cubes. For example, we write 5^2 for 25 and 5^3 for 125.

In general terms, an expression that represents repeated multiplication of the same factor is called a **power**. The number 5 is called the **base** and the number 2 is called the **exponent** (more often called as power). The exponent corresponds to the number of times the base is used as a factor.



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